

Week 6

Q1. Write a C++ Program to Add Two Numbers Using a Pointer.

```
#include <iostream>
using namespace std;

int main() {
    int a, b, sum;
    int *p1, *p2;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    p1 = &a;
    p2 = &b;

    sum = *p1 + *p2;

    cout << "Sum = " << sum << endl;

    return 0;
}
```

```
Enter two numbers: 5 7
Sum = 12

-----
Process exited after 6.49 seconds with return value 0
Press any key to continue . . .
```

Q2. Write a C++ Example Program for swapping numbers Using Pointers.

```
#include <iostream>
using namespace std;

void swapNumbers(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}

int main() {
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "Before swapping: a = " << a << ", b = " << b << endl;

    swapNumbers(&a, &b);

    cout << "After swapping: a = " << a << ", b = " << b << endl;

    return 0;
}
```

```
Enter two numbers: 8 12
Before swapping: a = 8, b = 12
After swapping: a = 12, b = 8

-----
Process exited after 11.75 seconds with return value 0
Press any key to continue . . . |
```

Q3. Write a C++ Program to print the address of the Variable Using a Pointer.

```
#include <iostream>
using namespace std;

int main() {
    int num;
    int *ptr;

    cout << "Enter a number: ";
    cin >> num;

    ptr = &num;

    cout << "Value of variable = " << num << endl;
    cout << "Address of variable = " << ptr << endl;

    return 0;
}
```

```
Enter a number: 153
Value of variable = 153
Address of variable = 0x6ffe34

-----
Process exited after 6.744 seconds with return value 0
Press any key to continue . . . |
```

Q4. Write a C++ Program for Increment and Decrement an Integer Using a Pointer.

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter number: " ;
    cin >> num;
    int *p = &num;
    (*p)++;
    cout << "After increment: " << *p << endl;
    (*p)--;
    (*p)--;
    cout << "After two decrements: " << *p << endl;
    return 0;
}
```

```
Enter number: 236
After increment: 237
After two decrements: 235

-----
Process exited after 7.669 seconds with return value 0
Press any key to continue . . . |
```

Q5. Write a C++ Program to Print a String Using a Pointer.

```
#include <iostream>
using namespace std;

int main()
{
    string str;

    cout << "Enter a string: ";
    getline(cin, str);

    string *ptr = &str;

    cout << "String: " << *ptr;

    return 0;
}
```

```
Enter a string: hello world
String: hello world
-----
Process exited after 18.07 seconds with return value 0
Press any key to continue . . .
```

Q6. Write a C++ program to concatenate two strings using pointers.

```
#include <iostream>
using namespace std;

int main() {
    char str1[200], str2[100];
    char *p1, *p2;

    cout << "Enter first string: ";
    cin.getline(str1, 100);

    cout << "Enter second string: ";
    cin.getline(str2, 100);

    p1 = str1;
    p2 = str2;

    while (*p1 != '\0') {
        p1++;
    }
    while (*p2 != '\0') {
        *p1 = *p2;
        p1++;
        p2++;
    }
    *p1 = '\0';
    cout << "Concatenated string = " << str1 << endl;

    return 0;
}
```

```
Enter first string: hello wo
Enter second string: rld
Concatenated string = hello world
-----
Process exited after 17.2 seconds with return value 0
Press any key to continue . . .
```

Q7. Write a program for reading elements using a pointer into an array and display the values using an array. i. Declare a set of elements. ii. Declare the pointer and initialize it to the first element's address of a set of elements(array). lii. Repeat the loop until the pointer reaches to the last element and displays each element.

```
#include <iostream>
using namespace std;

int main() {
    int arr[5];
    int *ptr;

    ptr = arr;

    cout << "Enter 5 elements: ";
    for (int i = 0; i < 5; i++) {
        cin >> *(ptr + i);
    }

    cout << "Array elements are: ";
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }

    cout << endl;

    return 0;
}
```

```
Enter 5 elements: 1 2 4 8 13
Array elements are: 1 2 4 8 13
```

```
-----
Process exited after 14.08 seconds with return value 0
Press any key to continue . . .
```

Q8. Write a program using a pointer variable to the sum of n elements from the array.

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    int arr[100];
    int *ptr;

    cout << "Enter the number of elements: ";
    cin >> n;

    ptr = arr;

    cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++) {
        cin >> *(ptr + i);
    }

    for (int i = 0; i < n; i++) {
        sum = sum + *(ptr + i);
    }

    cout << "Sum = " << sum << endl;

    return 0;
}
```

```
Enter the number of elements: 5
Enter 5 elements: 1 2 4 5 9
Sum = 21
```

```
-----
Process exited after 13.65 seconds with return value 0
Press any key to continue . . .
```

Q9. Write a program for reading elements using a pointer into the array and display the values using an array.

```
#include <iostream>
using namespace std;

int main()
{
    int n;

    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];
    int *ptr = arr;

    cout << "Enter " << n << " elements:" << endl;

    for (int i = 0; i < n; i++)
    {
        cin >> *(ptr + i);
    }

    cout << "Array elements are:" << endl;

    for (int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }

    return 0;
}
```

```
Enter the number of elements: 5
Enter 5 elements:
9 8 7 6 5
Array elements are:
9 8 7 6 5
-----
Process exited after 14.96 seconds with return value 0
Press any key to continue . . . |
```

Q10. Write a C++ program to reverse a string using pointers.

```
#include <iostream>
#include <cstring>
using namespace std;

int main()
{
    char str[100];

    cout << "Enter a string: ";
    cin.getline(str, 100);

    char *start = str;
    char *end = str + strlen(str) - 1;

    while (start < end)
    {
        char temp = *start;
        *start = *end;
        *end = temp;

        start++;
        end--;
    }

    cout << "Reversed string: " << str;

    return 0;
}
```

```
Enter a string: hello world
Reversed string: dlrow olleh
-----
Process exited after 10.7 seconds with return value 0
Press any key to continue . . . |
```